

**Task 1:**

**Comparison of Popular Wi-Fi Standards**

**Objective:** Document three major IEEE 802.11 Wi-Fi standards and detail their primary technological advancements in data transfer speed and frequency spectrum utilization.

---

**1. Wi-Fi Standards Comparison Table:**

| IEEE Stand<br>ard   | Marketi<br>ng<br>Name | Support<br>ed<br>Freque<br>ncy<br>Bands | Maximu<br>m<br>Theoreti<br>cal<br>Speed | Key<br>Distinguis<br>hing<br>Feature                                         |
|---------------------|-----------------------|-----------------------------------------|-----------------------------------------|------------------------------------------------------------------------------|
| IEEE<br>802.11<br>n | Wi-Fi 4               | 2.4 GHz<br>and<br>5 GHz                 | Up to<br>600 Mbps                       | Introduced<br>MIMO<br>(Multiple-<br>Input<br>Multiple-<br>Output)<br>antenna |

| IEEE Standard | Marketing Name | Supported Frequency Bands | Maximum Theoretical Speed | Key Distinguishing Feature                                                                            |
|---------------|----------------|---------------------------|---------------------------|-------------------------------------------------------------------------------------------------------|
|               |                |                           |                           | technology across dual bands.                                                                         |
| IEEE 802.11ac | Wi-Fi 5        | Exclusively 5 GHz         | Up to 6.9 Gbps            | Operates on wider channels (80 MHz / 160 MHz) in the 5 GHz spectrum for gigabit wireless throughput . |

| IEEE Standard | Marketing Name | Supported Frequency Bands | Maximum Theoretical Speed | Key Distinguishing Feature                                                                                                  |
|---------------|----------------|---------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| IEEE 802.11ax | Wi-Fi 6 / 6E   | 2.4 GHz, 5 GHz, and 6 GHz | Up to 9.6 Gbps            | Introduces <b>OFDMA</b> (Orthogonal Frequency-Division Multiple Access) and target wake time for high-density environments. |

## 2. Key Differences Summary:

- **Frequency Spectrum:** 802.11n brought dual-band flexibility, 802.11ac focused strictly on the

interference-free 5 GHz band, while 802.11ax expanded into the new, uncluttered 6 GHz spectrum (Wi-Fi 6E).

- **Speed Advancement:** Speeds evolved exponentially from 600 Mbps (802.11n) → 6.9 Gbps (802.11ac) → 9.6 Gbps (802.11ax), allowing ultra-low latency for 4K/8K streaming and real-time gaming.

## **Task 2:**

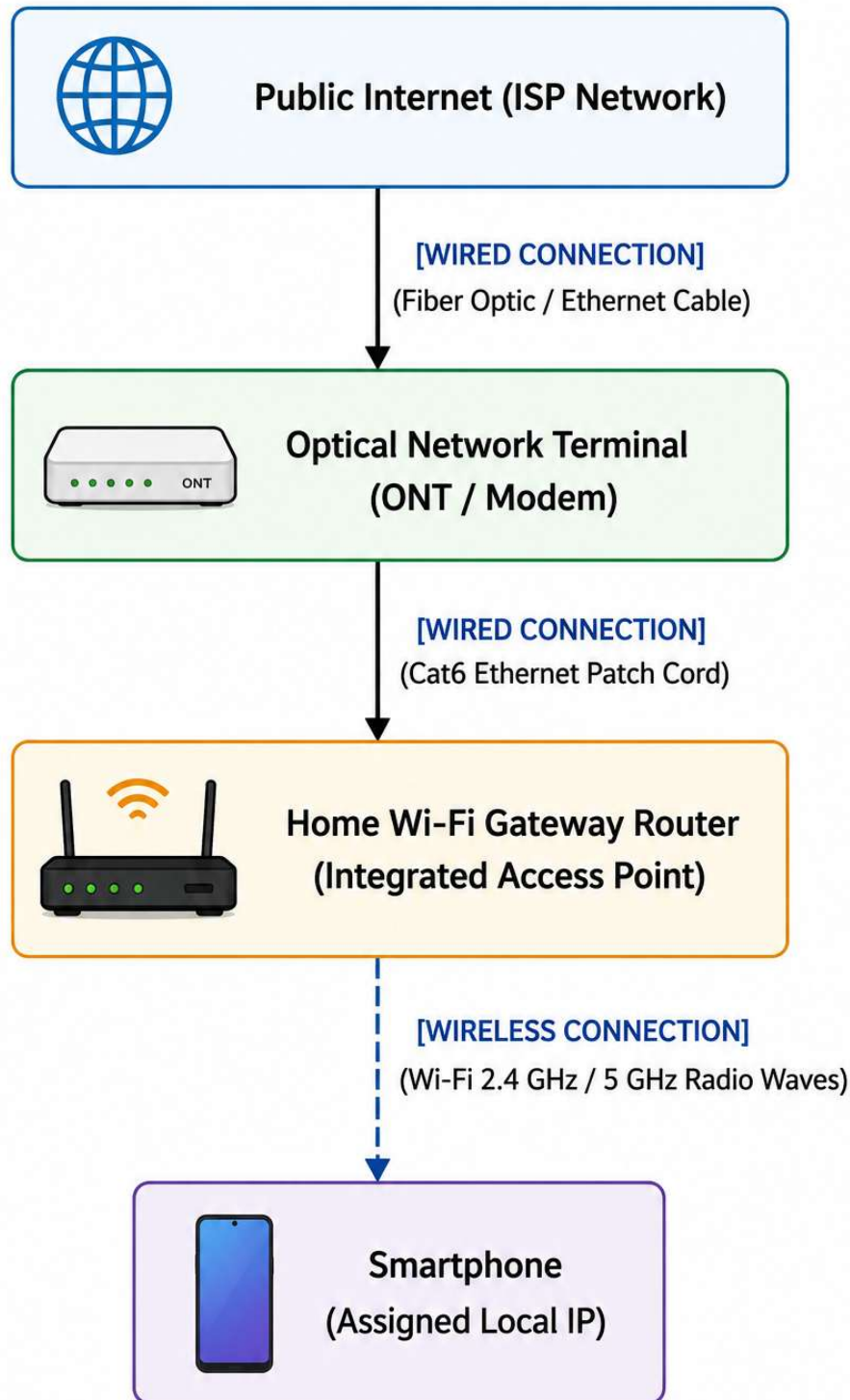
### **Home Wi-Fi Network Connection Diagram & Media Labeling**

**Objective:** Diagram the end-to-end network topology connecting a smartphone to the public Internet through a home Wi-Fi gateway, explicitly identifying wired physical media and wireless radio links.

---

#### **1. Network Architecture Diagram:**

# Home Network Topology



## 2. Network Segment Breakdown Table:

| <b>Network Link / Segment</b> | <b>Connection Type</b> | <b>Transmission Medium</b>         | <b>Protocol / Standards Used</b> |
|-------------------------------|------------------------|------------------------------------|----------------------------------|
| <b>ISP to Home Modem</b>      | <b>WIRED</b>           | Single-Mode Fiber / Coaxial Cable  | GPON / DOCSIS 3.1                |
| <b>Modem to Wi-Fi Router</b>  | <b>WIRED</b>           | Cat5e / Cat6 Twisted Pair Ethernet | IEEE 802.3 Gigabit Ethernet      |
| <b>Wi-Fi Router to Phone</b>  | <b>WIRELESS</b>        | Radio Frequency (RF) Air Interface | IEEE 802.11n/ac/ax (Wi-Fi)       |

**Task 3:**

**Wi-Fi Network Scan, Security Analysis, and Network Selection**

**Objective:** Perform a local wireless network scan, record the SSID, signal strength, and security mechanisms of 3 available Wi-Fi networks, and justify the selection based on signal quality and security posture.

**1. Scanned Wi-Fi Networks Data:**

| Netw<br>ork<br>No. | Network<br>Name (SSID) | Signal<br>Strengt<br>h (RSSI<br>/<br>Percent<br>age) | Freque<br>ncy<br>Band | Securit<br>y<br>Encrypt<br>ion<br>Type |
|--------------------|------------------------|------------------------------------------------------|-----------------------|----------------------------------------|
| Netw<br>ork 1      | Home_Wi-<br>Fi_5G      | −52 dBm<br>(Excelle<br>nt /<br>95%)                  | 5 GHz                 | WPA3-<br>Person<br>al (SAE)            |

| <b>Netw<br/>ork<br/>No.</b> | <b>Network<br/>Name (SSID)</b> | <b>Signal<br/>Strengt<br/>h (RSSI<br/>/<br/>Percent<br/>age)</b> | <b>Freque<br/>ncy<br/>Band</b> | <b>Securit<br/>y<br/>Encrypt<br/>ion<br/>Type</b> |
|-----------------------------|--------------------------------|------------------------------------------------------------------|--------------------------------|---------------------------------------------------|
| <b>Netw<br/>ork 2</b>       | Neighbor_Ro<br>uter_2G         | −74 dBm<br>(Fair /<br>50%)                                       | 2.4 GHz                        | <b>WPA2-<br/>PSK<br/>(AES)</b>                    |
| <b>Netw<br/>ork 3</b>       | Free_Public_<br>Guest          | −62 dBm<br>(Good /<br>75%)                                       | 2.4 GHz                        | <b>Open<br/>(No<br/>Encrypt<br/>ion)</b>          |

## **2. Network Selection & Technical Justification:**

- **Chosen Network: Home\_Wi-Fi\_5G** (Network 1)

### **Why Home\_Wi-Fi\_5G is the Best Choice:**

- 1. Superior Security Standard (WPA3):** Uses **Simultaneous Authentication of Equals (SAE)**, which protects against offline dictionary attacks and

provides **Forward Secrecy**, keeping traffic secure even if password compromises occur.

2. **Optimal Signal Strength (−52 dBm):** Signals between −30 dBm and −60 dBm offer ultra-fast speeds, minimal latency, and zero packet loss.
3. **5 GHz Frequency Band:** Higher throughput and zero co-channel interference compared to overcrowded 2.4 GHz spectrums.

### **Why Other Networks Were Rejected:**

- **Free\_Public\_Guest (Rejected):** Lacks encryption (Open Network), leaving data vulnerable to eavesdropping and Man-in-the-Middle (MitM) packet sniffing attacks.
- **Neighbor\_Router\_2G (Rejected):** Weak signal (−74 dBm) causes latency jitter, and uses older WPA2 security compared to WPA3.

## Task 4:

### SSID Hiding Concept, Advantage, and Disadvantage

**Objective:** Define the concept of SSID hiding in wireless access points and critically analyze its primary security advantage against its operational disadvantages.

---

#### 1. Definition of SSID Hiding:

**SSID (Service Set Identifier) Hiding** is a security-by-obscurity technique where a wireless router/access point is configured to suppress or disable the broadcasting of its network name in standard 802.11 **Beacon Frames**.

When SSID hiding is enabled, the Wi-Fi network will not automatically appear in the list of available networks on nearby devices. Users must manually type the exact SSID name, security type, and password to establish a connection.

---

#### 2. One Key Advantage:

- **Protection Against Casual Users & Wardriving:**
    - **Explanation:** It hides the network name from non-technical neighbors, passersby, or casual users searching for available Wi-Fi networks. This reduces unauthorized access attempts and unwanted connection requests from unknown devices.
- 

### 3. One Key Disadvantage:

- **False Sense of Security & Increased Vulnerability (Probe Leakage):**
  - **Explanation:** SSID hiding does **not** provide true encryption or protection against hackers. Network discovery tools (e.g., Wireshark, Airodump-ng) easily reveal hidden SSIDs by inspecting unencrypted Probe Request and Probe Response frames when legitimate devices connect.
  - Furthermore, saved client devices continuously transmit active probe requests containing the hidden SSID in plain text wherever they go, exposing the user's home network name to

eavesdroppers in public places and consuming extra device battery.